DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular & Supplementary Winter Examination-2023

Course: B. Tech. Branch : CSE(AI&ML)/AI&DS/AI Semester :V

Subject Code & Name: BTAIC502 Machine Learning

Max Marks: 60 Date:03-01-24 Duration: 3 Hr.

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.**		**12**
A)	Define Machine Learning. Describe the application of Machine Learning	**Understand**	**6**
B)	Compare and contrast the Classification, Regression and Clustering on the basis of following Parameter	**Analyze**	**6**

- Type
- Basic
- Includes anticipation of
- Characteristics of anticipated data
- Procedure for calculating
- Algorithms

C)	Explain the types of Machine Learning with suitable Diagram and Example	**Understand**	**6**
Q.2	**Solve Any Two of the following.**		**12**
A)	Explain the Confusion Matrix with example and state the nedd of confusion matrix	**Understand**	**6**
B)	For the given confusion matrix Calculate	**Apply**	**6**

N=200	**Prediction = No**	**Prediction = Yes**
Actual = No	60	10
Prediction = Yes	5	125

1. Accuracy
2. Error Rate
3. Precision
4. Recall
5. Specificity
6. F1 Score

C)	Explain the Performance Metrics for Regression Problems- Mean Absolute Error (MAE), Mean Square Error (MSE), R Squared (R2) And for the following Table Calculate 1. MAE 2. MSE 3. RMSE	**Apply**	**6**

Ticket #	Actual Price	Predicted Price
1	250	265
2	110	140
3	500	480
4	200	215
5	330	290
6	490	515
7	670	750
8	210	210
9	435	420
10	375	285

Q. 3	**Solve Any Two of the following.**		**12**
A)	What is logistic regression? Explain the steps in Logistic Regression Learning. Also give the cost function of Logistic Regression	**Understand**	**6**
B)	Explain the Gradient Descent and how it helpful to maximize the learning rate in regression	**Understand**	**6**
C)	A. The values of y and their corresponding values of y are shown in the table below	**Apply**	**6**

x	0	1	2	3	4
y	2	3	5	4	6

a) Find the least square regression line y = B1 x + B0
b) Estimate the value of y when x = 10.

Q.4	**Solve Any Two of the following.**		**12**
A)	Consider the following dataset S in table for question where each record represents the age, income and is a student or not. And we need to classify it as buyers/non-buyers of computer.		**6**

rec	Age	Income	Student	Credit _rating	Buys_computer
r1	<=30	Hight	No	Fair	No
r2	<=30	Hight	No	Excellent	No
r3	31...40	Hight	No	Fair	Yes
r4	>40	Medium	No	Fair	Yes
r5	>40	Low	Yes	Fair	Yes
r6	>40	Low	Yes	Excellent	No
r7	31...40	Low	Yes	Excellent	Yes
r8	<=30	Medium	No	Fair	No
r9	,=30	Low	Yes	Fair	Yes
r10	>30	Medium	Yes	Fair	Yes
r11	<=30	Medium	Yes	Excellent	Yes
r12	31...40	Medium	No	Excellent	Yes
r13	31...40	High	Yes	Fair	Yes
r14	>40	Medium	No	Excellent	No

1. Calculate the information Gain(S,income)?
2. Calculate the information Gain(S,Credit_score)?
3. Calculate the information Gain(S,student)?

B) Explain Overfitting and Pruning with respect to Decision Tree **Understand** **6**

C) Explain the Random forest Algorithm with examples. **Understand** **6**

Q. 5 **Solve Any Two of the following.** **12**

A) For the following Dataset use Naïve Bayes Classifier to classify following problem **6**

Example No.	Color	Type	Origin	Stolen?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

Classify

Color	Type	Origin	Stolen
Red	SUV	Domestic	?

B) Explain Kernel Trick in SVM? Kernel Trick is helpful to solve non-linearly separable problem in SVM. Justify **Understand** **6**

C) Explain the working of KNN Algorithm **Apply** **6**

For the Given Dataset use to Classify Dog, Pig and Cat with given features.

Height	Width	Class
0.1	0.1	cat
0.2	0.5	cat
0.3	0.2	cat
0.4	0.3	cat

0.5	0.6	cat
0.6	0.1	Pig
0.7	0.2	Pig
0.8	0.5	Pig
0.9	0.3	Pig
0.6	0.9	Dog
0.7	0.5	Dog
0.8	0.6	Dog
0.9	0.7	Dog

Use 3-NN and 5 NN to find the Class for following unknown data Use Euclidian Distance as Distance metric for given dataset

Height	**Width**	**Class**
0.6	**0.6**	**???**

***** End *****